# Darwin and Evolution I

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**Evolution:** a theory describing the processes that have transformed life on Earth from its earliest forms to the great diversity of organisms existing today.

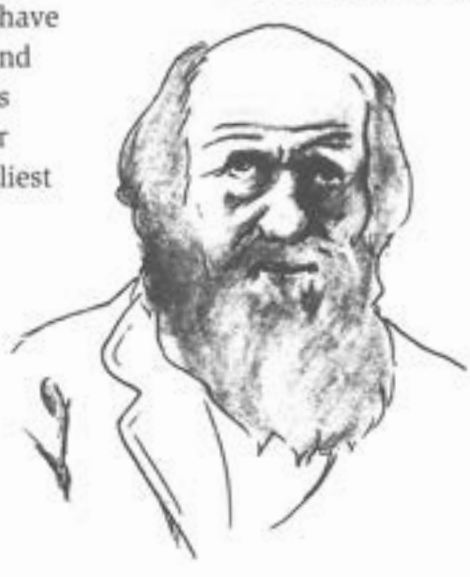
The vast majority of scientists use the *Theory of Evolution* to explain how the species of plants and animals that live (and have lived) on the Earth have come about. By understanding why species have evolved, scientists can predict, with some accuracy, how outside factors might affect future generations. The theory also helps scientists to better understand relationships between fossils and to learn more about our past.

1 Explain what is meant by the term 'fossil'.

Naturalists and scientists have long believed that species changed, but were uncertain as to why the changes occurred.

2 'A species can be described as a group of similar organisms that can breed with each other.' True or false?

Charles Darwin (1809–82) questioned many things, such as why some insects were rare and others were common. He observed that cows introduced to graze in a field applied a *directional pressure* on grasses growing where they grazed. Although the whole area may have started with an identical mix of, for example, quick growing and slow growing grasses, the ratio of grasses began to change if cows were present. This was because the cows selectively ate (or ignored) certain species of grass. This was one of the earliest documented examples of *selective pressure*.



3 Which type of grass would do better in a field grazed by cows, a quick-growing grass or a slow-growing grass? Explain your answer.

4 Suggest one other factor that might influence the ratio of grasses present if cows were present.

5 In your own words, describe what is meant by 'selective pressure'.

6 Give another example of selective pressure.

Darwin was a very observant person and he was the first to suggest that the pressures of nature could affect how species changed over time.

7 Write the name of a common plant or animal and give three examples of natural pressures that could affect this species' ability to grow and reproduce.

8 Write the name of a common plant or animal and give three examples of directional pressures, applied by humans, that could affect this species' ability to grow and reproduce.

Darwin believed that some individuals in a population possessed characteristics slightly different from those of other members. These differences might better enable some individuals to survive than the rest. Those who were better equipped to compete in their surroundings survived and reproduced in greater numbers. The offspring ('children') of the successful parents inherited the traits (characteristics) that made the previous generation better survivors. Because more young of successful parents were born, the percentage of the population possessing these traits increased. It is possible that Darwin was influenced by his grandfather's book, *Zoonomia*. This book, which wasn't a scientific text but more a book of musings, had a central theme of how the environment affects an organism's survival.

9 Give a brief definition of 'environment'.



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